Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
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Project Title:	Boron-Doped Microcrystalline and Nanocrystalline Diamond Thin-Films: New Materials for Electrochemical Detection in Microfluidic Systems and Electrolytic Water Purification		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
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Comments:			
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No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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COI Name (Institution):			
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Research during FY02 was directed toward developing electrochemical-based water quality monitoring methods for Advanced Environment Monitoring and Control (AEMC) using electrically conducting diamond thin film. Diamond is a new electrode material that outperforms other common electrodes (e.g., glassy carbon) by providing significantly improved analytical figures of merit (e.g., linear dynamic range, limit of detection, response precision, and response stability) and, in some cases, enabling the detection of analytes that cannot be easily monitored with other electrodes. Our goal is to create assays that are easy to use, function in a microgravity environment, small in scale, reliable, sensitive, and stable.
In one task, we are using the diamond electrodes for the detection of trace metal ions (e.g., Ag(I), Cu(II), Cd(II), Pb(II), and Zn(II)) via anodic stripping voltammetry (ASV). ASV involves a two-step measurement: (i) electroreducing the

Task Description:	metal ions at a negative potential to form metal deposits on the electrode surface, thereby preconcentrating the analytes and (ii) selectively electrooxidizing each metal during a slow potential scan in toward positive potentials. Hg and Hg-coated electrodes are often used for ASV; however, there is a need to develop alternate electrodes that possess the same attractive properties as Hg, because of toxicity and volatility issues. Diamond possesses many of the same attributes as Hg and, therefore, appears to be a viable material for this electroanalytical measurement. Diamond exhibits superb properties for the detection of metal ions at low ppb levels. We demonstrated that the electrode could be used to accurately determine the concentrations of metal ions in city drinking water, waste treatment sludge and contaminated soil. Work in FY03 will focus on implementing pulsed electrodeposition for the preconcentration step, miniaturizing the electrochemical cell and standardizing the steps in the analysis, and applying the method to the analysis of water samples from the International Space Station (ISS). In a second task, diamond electrodes are being combined with electrophoretic separation in a microchip format (so-called laboratory-on-a-chip) for the separation and detection of chemical contaminants in water supplies. Plastic microchips are being used (e.g., PDMS and PMMA). Diamond is unique in the carbon electrode family in that is possesses the requisite properties to support anodic oxygen transfer reactions. Important biological and environmental analytes that undergo oxidation via anodic oxygen transfer include amino acids, indoles, alkylhydrazines, chlorinated phenols and alkanolamines. We are focusing presently on the electrophoretic separation and amperometric detection of alkanolamines. These compounds are used extensively by the chemical and pharmaceutical industries as lubricants, corrosion inhibitors and ingredients in various pharmaceutical preparations. They have been identified as pollutants and there is a strong need to quantify alkanolamines with high sensitivity and accuracy, and without extensive sample manipulation. This class of analytes is useful for developing the diamond/microchip analytical methodology and, more importantly, is a reasonable mimic for antibiotics in water - an emerging issue in water quality monitoring. We constructed an appropriate cell holder and sample reservoirs for the diamond/microchip assembly, constructed and tested a multichannel power supply for controlling the sample injection and separation voltages, and demonstrated that the chips can be effectively sealed and that diamond microelectrodes can be reproducibly positioned at the end of the separation channel for electrochemical detection. We completed a preliminary cyclic voltammetric investigation of the electrooxidation of three model alkanolamines (3-aminopropanol, 4-aminobutanol and 5-aminopentanol) in an alkaline electrolyte solution, as a function of the potential scan rate, scan number and analyte concentration. We have also established the appropriate conditions for electrophoretically separating these analytes in a fused silica capillary column. Work in FY03 will focus on electrophoretically separating these analytes on the microchip and amperometrically detecting them with diamond microelectrodes. A complete set of analytical figures of merit will be obtained. We also plan to exand the scope of the work to include the separation and detection of alkylhydrazines. In a third task, we are exploring the use of diamond anodes for the electrochemical treatment of wastewater. Diamond functions as a dimensionally stable anode and is extremely stable at high current densities and in harsh chemical environments, exhibits a large overpotential for oxygen evolution in aqueous media, and supports anodic oxygen transfer reactions. This new electrode material appears to be quite useful for the remediation of organic and inorganic waste. We demonstrated the use of a diamond anode for the oxidative remediation of atrazine. Work in FY03 will focus on testing the effectiveness of the electrode at remediating simulated wastewater from a space crew.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The main goal of this research is to identify and develop advanced electroanalytical methods that can benefit the NASA Advanced Environmental Monitoring and Control (AEMC) program. The technologies being developed can be used to detect both biologically and environmentally-important analytes with high sensitivity, good precision and stability, and with minimal user input. While these advanced analytical methods are useful for space applications (light weight, low volume, superb figures of merit), they are also highly practical for water quality monitoring on Earth. The use of diamond anodes for the electrochemical treatment of wastewater could become widespread in both space and on earth, particularly after more is learned about the effectiveness (e.g., conversion rate, conversion efficiency, completeness of the oxidation) of the electrode at remediating different forms of organic and inorganic waste.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 08/03/2009)